

VISHOD - Visual Dataset Scraping and Hybrid Outlier Detection

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Abstract—In supervised learning, the effectiveness of classification models is intrinsically linked to the quality of the underlying dataset. Nonetheless, the construction of large-scale, accurately labeled image datasets remains a resource-intensive and time-consuming endeavor—particularly in contexts where offline operation and limited computational resources are critical constraints. This paper presents an automated method for building and refining image datasets using web scraping and a hybrid outlier detection approach. The proposed pipeline comprises four main stages: (i) automated image collection from online sources; (ii) extraction of semantic representations using deep learning models; (iii) detection and elimination of outlier samples through a hybrid, consensus-driven anomaly detection approach; and (iv) dataset validation using both visual inspection and statistical performance indicators. As a case study, we develop a lightweight image classifier for recognizing common two-dimensional geometric shapes, aimed at supporting educational applications in contexts with limited internet connectivity, such as the Amazon region. The results show that the method allows the construction of balanced, high-quality datasets without manual annotation, with a 25.6% increase in the average PCA variance, a 5.6% reduction in the average distance to the centroid, and a 21.6% increase in the average structural similarity index of the dataset after the process, evidencing greater diversity, cohesion, and visual homogeneity in the data. In addition, this study demonstrates that the proposed method is a robust solution for building and cleaning datasets, particularly suitable for resource-limited environments that require customized and domain-specific data.

Index Terms—Dataset Curation, Web Scraping, Outlier Detection, Image Classification, Deep Learning.

I. INTRODUCTION

Image classification is a fundamental technique in areas such as robotics [1], Computer-Aided Design (CAD) [2], and medical image diagnosis [3], among other industrial applications [4]. It allows automated systems to identify and classify objects or patterns in images [5], facilitating real-time decision-making. In robotics, image classification helps robots understand their surrounding environment and execute actions based on visual recognition [6]. In CAD, it enhances

the precision and efficiency of design and production processes, optimizing everything from model creation to quality control [7]. In medical image diagnosis, image classification is used for, for example, bone metastasis, as evidenced in [8]. The effectiveness of image classification models critically depends on the availability of extensive and properly labeled datasets, specific to the desired application ([9], [10], [11]).

Obtaining large volumes of labeled data to train classification models is an inherently complex, manual, and costly process [12]. This task not only requires significant time and effort but also demands high-quality, representative data to enhance model learning [13]. This challenge, coupled with the exponential growth of visual data, positions Automatic Image Annotation (AIA) approaches as fundamental to this process. This is evidenced by the study [12], which reinforces the problem and categorizes AIA techniques into five architectures that differ in design, feature extraction, precision, complexity, and performance. Although these architectures have led to significant advances in AIA, important challenges persist, such as the analysis of high-dimensional features, the semantic gap between images and labels, and the scarcity of comprehensively annotated datasets [14].

Although promising, AIA-based approaches face significant limitations when applied to specific domains. As discussed by Whang et al. (2023), much of the automatically collected data is noisy, biased, or even contaminated by malicious instances (e.g., data poisoning), which compromises the effectiveness of the trained models. Furthermore, many of these methods do not systematically integrate curation, cleaning, or outlier detection steps, which compromises the final quality of the dataset and, consequently, the performance of the trained model [16]. Thus, there remains a need for more integrated and targeted solutions capable of automating not only data collection but also its organization and refinement, with a focus on computationally efficient and semantically demanding tasks, such as classifying specific shapes and everyday objects.

Given the limitations of traditional and automated methods for obtaining specific data, web scraping emerges as an effective approach to address the scarcity of information targeted for particular tasks [17]. This technique relies on using scripts and algorithms that can navigate web pages and systematically extract content, enabling the construction of large, diverse, and thematically oriented databases [18]. By automating data collection, web scraping significantly reduces the required manual effort while also offering scalability and a high degree of customization—essential characteristics for applications that demand semantic control over visual data [19]. Such attributes

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are particularly valuable in domains like pattern recognition and machine learning, where the quality and quantity of available data play a decisive role in the performance of computational models [20].

Despite the advantages of web scraping in obtaining large volumes of visual data, images extracted directly from the web frequently exhibit quality issues, such as noise, semantic inconsistencies, and outliers [21]. This noise can manifest as irrelevant images, low resolution, visual distortions, or contexts that do not adequately represent the element of interest. Such imperfections compromise the ability of machine learning models to learn significant patterns, leading to generalization errors and performance degradation [22]. Therefore, it is crucial to adopt robust preprocessing and filtering mechanisms to ensure the consistency, representativeness, and effectiveness of the constructed dataset.

Data diversity is a critical factor that can impact the uniformity and relevance of the collected images. The diversity of images from various sources can lead to significant variations in terms of resolution, framing, lighting, visual style, and context [23]. Although this diversity can, in some cases, contribute to the model's robustness, it also tends to compromise the uniformity and representativeness of the dataset. Furthermore, the lack of control over the captured content can result in the inclusion of identical or ambiguous images, which hinders the learning process and can negatively affect the models' generalization [24]. These factors reinforce the need for automated refinement strategies that not only filter out noise and outliers but also ensure the coherence and controlled diversity of the dataset.

According to Lotfi et al. (2021), the lack of standardization is a significant challenge when using web scraping to collect data, especially for specific applications like shape classification. There are no widely established methods to ensure the consistency and quality of images extracted from the web, which can result in disorganized and inconsistent datasets [26]. The absence of a clear standard for image labeling and curation hinders the creation of datasets that can be easily used in different contexts or models [27]. Furthermore, variation in data sources can lead to discrepancies in image formats, resolutions, and metadata, making it more difficult to ensure that all collected images meet the specific requirements of a classification task. This increases the risk of including irrelevant or mislabeled images in the dataset, thereby compromising the training and generalization of the models.

In addition to the challenges posed by source diversity, the lack of standardization in collection and curation processes represents a significant obstacle to the efficient use of images obtained via web scraping [28]. Currently, there are no widely established guidelines to ensure the semantic consistency, quality, or contextual appropriateness of extracted data, which often results in disorganized and inconsistent datasets. The lack of a clear protocol for labeling, organization, and validation hinders the reusability of this data in different experimental scenarios and compromises interoperability between models [29]. In contexts requiring lightweight and efficient models, such as mobile or embedded applications,

these gaps become even more critical, as each instance of noise or inconsistency can directly impact the system's precision and stability [30]. Thus, more than a technical barrier, the absence of standardization represents a strategic opportunity for the advancement of automated methods that combine flexibility with the structural rigor demanded by highly specialized domains.

To address the challenges related to the acquisition, standardization, and qualification of visual data for specific tasks, this work aims to develop an automated and structured method for building image datasets for classification tasks. The central focus is on creating an integrated approach that allows for efficiently collecting images, identifying and removing inconsistent or irrelevant data, and finally, evaluating the quality of the generated dataset based on objective criteria. The proposal aims to address the need for solutions that reconcile scalability, quality control, and low computational cost, enabling practical application in real-world scenarios. By systematically structuring this process, the work contributes to filling a gap in the literature by offering a replicable model for generating reliable data in specific contexts.

Based on the established objective, this work proposes a structured method composed of three main phases: (i) automated image collection via web scraping, described in Subsection IV-A, focusing on capturing diverse and representative visual content of two-dimensional geometric shapes; (ii) automated outlier detection and removal based on a hybrid approach, detailed in Subsection IV-C, which integrates techniques for structural analysis, efficient indexing, dense clustering, and anomaly detection; and (iii) quality assessment of the generated dataset, further detailed in Subsection IV-B, based on visual similarity metrics and statistical analysis. The proposal stands out for its integrated and sequential organization of these stages, resulting in a robust and computationally low-cost pipeline suitable for applications requiring standardized and semantically consistent data.

Data collection is intelligently structured through the combination of terms, such as visual descriptors and contextual modifiers, to formulate queries for search engines. The dataset's quality is evaluated using statistical methods like Principal Component Analysis (PCA) ([31], [32]) and Euclidean Distance [33], allowing for the measurement of the diversity and cohesion of the generated set. Additionally, the Structural Similarity Index Measure (SSIM) [34] is used to quantify the structural quality between image pairs. As pointed out by Torralba et al. (2011), the combined use of these metrics is effective in ensuring the integrity and representativeness of visual datasets for machine learning tasks. Finally, the automated detection and removal of outliers is a particularly critical part of the process, conducted through a combination of complementary techniques. Hierarchical Density-Based Spatial Clustering (HDBSCAN) [36] identifies non-linear clusters of inconsistent samples; Isolation Forest [37] detects anomalies based on isolation criteria; and Mahalanobis Distance [38] is employed to detect statistically discrepant instances based on the multidimensional distribution of the data.

To demonstrate the applicability of the proposed method

in a practical context, a case study was conducted centered on the development of an Artificial Intelligence (AI) model for the GeoMeta¹ mobile application. This application aims to support geometry education for elementary school students through the automatic classification of two-dimensional geometric shapes, such as squares, circles, triangles, and rectangles, found in everyday objects. However, the absence of specific public datasets for this task represents a significant barrier, as the manual collection and curation of images would require time, expertise, and high operational costs. Given this scenario, the method proposed here offers an automated, efficient, and low-cost solution for building a custom dataset, ensuring the representativeness and quality of the data used for training the classifier, while also meeting the requirements for offline operation and compatibility with low-performance mobile devices, as demanded by the realities of the Amazon region.

The application of the method in the proposed scenario followed the complete pipeline described previously. Initially, images were collected via web scraping using carefully structured search terms to represent nine distinct geometric classes associated with everyday objects: square, rectangle, triangle, circle, ellipse, parallelogram, trapezoid, pentagon, and hexagon. Subsequently, the data was automatically filtered through the detection and removal of redundant, inconsistent, or semantically irrelevant images, using the proposed hybrid approach. Finally, the dataset was evaluated based on quantitative metrics for visual quality and statistical dispersion, ensuring its suitability for training the classification model. From the validated dataset, a low-cost experimental computational model was developed, incorporating hyperparameter optimization, supervised training, and subsequent performance evaluation, with the goal of its integration into the educational application.

The main contributions of this work are: (i) the proposal of an automated and structured method for building image datasets for specific contexts, integrating collection, cleaning, and dataset quality assessment; (ii) the application of a hybrid and multidimensional approach for outlier detection, combining perceptual and statistical metrics; (iii) the demonstration of the method's applicability in a real-world educational scenario, geared towards building a mobile application to support geometry teaching; and (iv) the provision of a replicable, low-cost computational pipeline aimed at classification tasks in domains with a scarcity of labeled data. The remainder of this article is organized as follows: Section II presents related works and discusses gaps in the literature; Section III details the motivation and relevance of the proposed case study; Section IV describes the methodology employed in this work; Section V discusses the obtained results; finally, Section VI provides the conclusions and suggestions for future work.

II. RELATED WORKS

In [39], a technique is presented for detecting outliers in images using Poisson interpolation, a method traditionally

used in computer graphics for image editing. The proposed approach involves predicting pixel values in corrupted or missing data regions of an image using Poisson interpolation. Outliers are identified by comparing the actual pixel values with the predictive values generated by the interpolation. When a significant discrepancy exists between these values, an outlier is detected. The paper validates the technique through experiments, demonstrating its ability to accurately and efficiently detect outliers, thus contributing to enhancing the quality of image processing.

The work in [40] presents a systematic review of machine learning techniques applied to anomaly detection. The research analyzes machine learning models used to identify and extract anomalous data components in different contexts, a crucial task in various domains such as cybersecurity, system monitoring, and financial data analysis. The study covers the main approaches, including supervised, unsupervised, and semi-supervised methods, discussing their advantages, limitations, and areas of application. Furthermore, it presents current challenges in the field, such as the need to handle large volumes of data, the variability of anomaly patterns, and the scarcity of labeled data.

In [41], the quality of ultra-low-dose computed tomography (CT) images is evaluated using deep learning techniques for post-processing. The comparison between post-processing algorithms revealed that the vendor-independent method (DLM), especially with the standard kernel (DLM-std), achieved the best overall image quality. The DLM-std exhibited lower noise, fewer artifacts, and better spatial resolution compared to vendor-specific techniques (DLIR). Although DLIR showed better performance in terms of noise reduction, DLM-std was preferred due to its superior visual quality, featuring sharper images with fewer distortions.

The work in [42] proposes a data transformation to optimize the application of Euclidean distance in clustering analyses, especially on biomedical data. The technique adjusts variables to improve the separation between groups, minimizing their overlap and maximizing the intra-cluster distance. The Euclidean Distance Optimization (EDO) transformation improves clustering effectiveness by addressing the variability of data scales, showing superiority over traditional methods like z-standardization, particularly on genomic datasets for breast cancer. Euclidean distance is crucial in this process, as the transformation ensures that the distances between groups are better defined, leading to improved clustering results.

In [19], the methodology for collecting news related to COVID-19 via web scraping is described to create a dataset geared towards sentiment and emotion analysis. The research uses automated techniques to extract information from news websites and then processes this data to identify prevalent sentiments and emotions in the publications, aiming to understand how the media covers the pandemic. The study demonstrates how using web scraping can provide a valuable basis for research in sentiment analysis on large volumes of text, highlighting the relevance of such tools for understanding the emotional impact of the pandemic on people.

The work [43] presents an optimized implementation of the HDBSCAN* algorithm in the Tribuo machine learning

¹https://play.google.com/store/apps/details?id=com.Inteceleri.GeoMeta&hl=pt_BR

library, developed in Java. HDBSCAN* was used for advanced clustering on complex data, aiding in the identification of clusters, outliers, and providing rapid predictions. This enhanced version uses concurrency to improve performance and features a prediction functionality, allowing the HDBSCAN* model to make fast predictions on unseen data, which is valuable for applications like real-time clustering services. The implementation also highlights the importance of Euclidean distance in the calculation of core distances, which is fundamental for constructing the minimum spanning tree and forming the cluster hierarchy. This consideration of Euclidean distance ensures that proximity relationships between points are maintained, facilitating the identification of significant clusters and the detection of outliers.

The reviewed works present relevant approaches that touch upon aspects of this paper's proposal, but they differ in terms of integration and methodological scope. For example, the technique based on Poisson interpolation [39] focuses on detecting outliers directly in image pixels, whereas the present proposal uses feature vectors extracted by neural networks to perform a contextual multivariate analysis. The systematic review by Nassif et al. (2021) highlights the diversity of anomaly detection methods but does not propose integration mechanisms for data collection, filtering, and curation—central elements of our method. Similarly, although Thota et al. (2021) explore the application of web scraping to build datasets, their focus is on text and sentiment analysis, without considering images or the need for task-oriented curation. The optimized HDBSCAN implementation presented by Stewart et al. (2022) relates to this work through its use of the algorithm for outlier detection, but it is applied in a generic clustering context, without addressing its use in hybrid filtering processes for images. Thus, this paper differentiates itself by proposing an automated approach, guided by task-specific data filtering, that combines scraping, embedding extraction, and multiple outlier removal techniques for curating images intended for visual classification.

III. CASE STUDY

This case study aims to demonstrate the practical applicability of the developed methodology (Section IV), focusing on the automated classification of two-dimensional geometric shapes from everyday images. The central motivation for choosing this scenario stems from the need to optimize the educational application GeoMeta², a platform geared towards teaching geometry using immersive technologies, such as Augmented Reality (AR) and Virtual Reality (VR), in educational environments with limited technological infrastructure.

This need arises from one of the goals defined in the project approved under the Public Call³, related to the project titled "EduTech Amazon: Uso de inteligência artificial para reconhecimento de padrão geométrico em streaming em uma aplicação no ensino da educação básica através do App

Geometricando VR e AR" (EduTech Amazon: Use of artificial intelligence for geometric pattern recognition in streaming in an application for basic education teaching through the Geometricando VR and AR App)⁴. This project aims to enhance GeoMeta to its 2.0 version to facilitate the teaching and learning of basic education Mathematics, especially plane and spatial geometry, by using AI to recognize patterns in streaming within VR and AR environments created from objects and landscapes of the Amazon region.

GeoMeta was designed to promote a more interactive and meaningful learning of geometry, offering students visual and manipulable experiences of mathematical concepts, as previously discussed in earlier studies [44]. Figure 1 displays the application's interface in AR and VR, developed to detect geometric patterns in the real environment and associate them with the corresponding geometric shapes using AI. The figure shows the identification of the geometric shape sphere, corresponding to the detection of a soccer ball visualized in the scene. The application's operation, however, still relies on manual or pre-programmed interactions, lacking an autonomous mechanism for the efficient visual recognition of geometric shapes in the physical environment [45].

In this context, a low-cost computational model is proposed, capable of automatically identifying shapes such as triangles, squares, rectangles, and circles, among others, directly from images obtained via mobile device cameras. The choice of a lightweight and efficient model is justified by the reality of various locations in the Amazon region, where internet access is unstable or nonexistent, and the available devices have limited computational resources [46]. Evidence of this can be found in data published by the Regional Center for Studies for the Development of the Information Society (Cetic.br), which indicate that 21% of households in the Northern Region of Brazil, encompassing a large part of the Brazilian Amazon, do not have internet access [47].

To enable this solution, a complete process was conducted, involving the creation of a custom dataset containing images of real objects labeled according to their geometric shapes, followed by a meticulous outlier removal stage. The modeling stage was guided by principles of efficiency, employing deep learning techniques adapted for low-computational-capacity environments, with the goal of deploying the model directly on smartphones.

Thus, this case study exemplifies the applicability of the proposed method, showcasing its potential to expand the reach of GeoMeta, making it more inclusive, functional, and adapted to different educational realities, including in the Legal Amazon and contexts with limited infrastructure. The integration of the model into the application represents a significant advancement in the use of artificial intelligence as a pedagogical tool, promoting greater autonomy in learning and democratizing access to mathematical knowledge.

IV. METHODOLOGY

The methodology applied in this work is organized into four main stages, as illustrated in the blocks of Figure 2:

²https://play.google.com/store/apps/details?id=com.Inteceler.I.Geometa&hl=pt_BR

³CNPq/MCTI/SEMPI Call No. 21/2021 – Human Resources Training Program in Strategic Areas (RHAE)

⁴<https://www.youtube.com/watch?v=kZAXQxGE-oA>

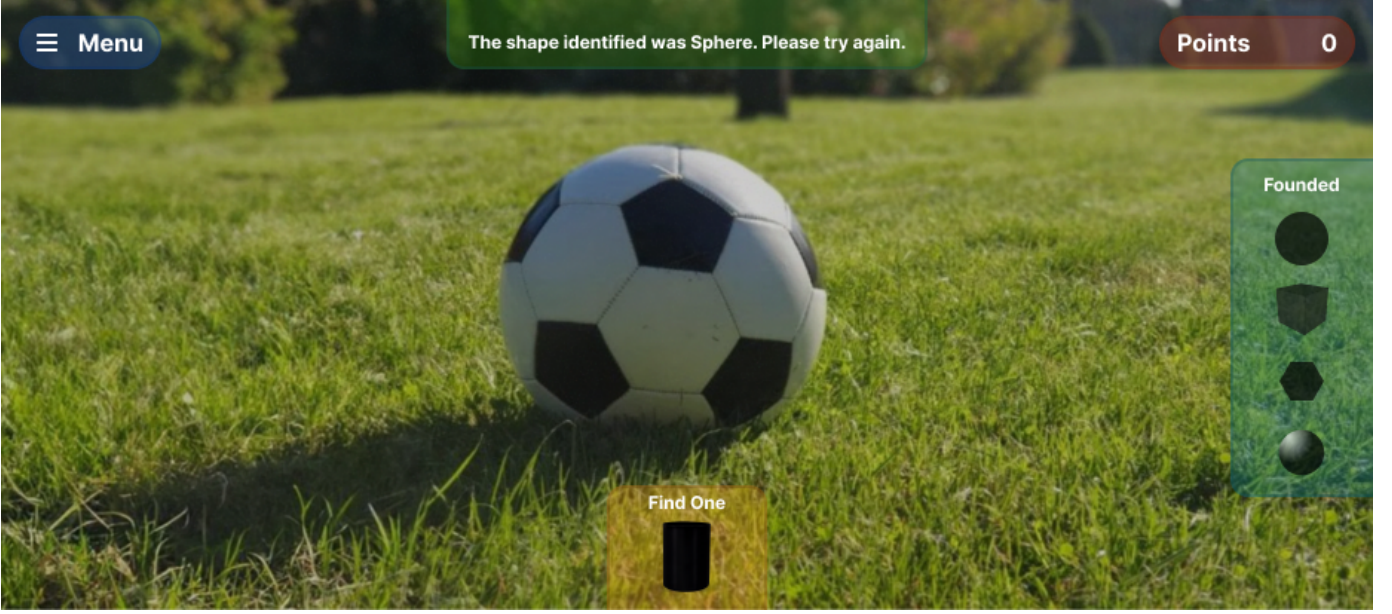


Fig. 1. Prototype of the AR module interface for the GeoMeta application.

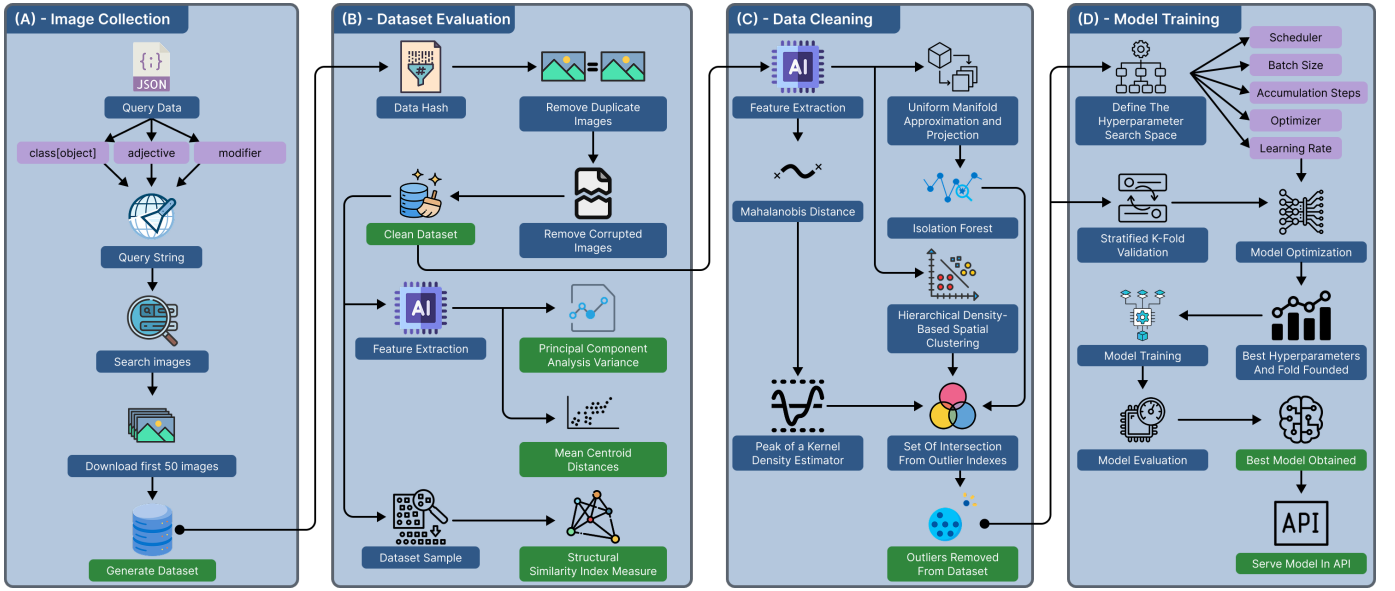


Fig. 2. Overview of the proposed methodological pipeline for the presented case study. The process is divided into four stages: (A) image collection via structured scraping, (B) automated dataset assessment, (C) outlier detection and removal using a hybrid approach, and (D) classifier training and validation.

(A) image collection, (B) dataset assessment, (C) data cleaning, and (D) classifier training. The first stage, described in Subsection IV-A, involved the automatic collection of images from the internet and the initial structuring of the dataset. In the dataset assessment stage, detailed in Subsection IV-B, duplicate and corrupted images were removed from the dataset, and its quality was measured. The third stage, further detailed in Subsection IV-C, consisted of the automatic cleaning of the dataset, in which outliers found through the union of multiple techniques were removed. Finally, Subsection IV-D describes the final stage of the classifier training methodology, in which an experimental classification model was optimized and trained on the dataset generated in this case study.

A. Image Collection

The construction of the dataset for classifying geometric shapes in real-world images required a systematic collection methodology, spanning from the definition of search terms to the structured storage of the images. This section details the process in four main stages: (i) definition of search data, (ii) linguistic structuring of queries, (iii) implementation of web scraping, and (iv) dataset organization.

The parameters for constructing the queries were organized into four main elements: (i) classes, divided into square, rectangle, triangle, circle, ellipse, parallelogram, trapezoid, pentagon, and hexagon; (ii) descriptive adjectives (e.g., color, texture, physical state); (iii) contextual modifiers, such as

TABLE I
EXAMPLES OF COMBINATIONS OF TERMS USED TO GENERATE SEARCH STRINGS IN THE WEB SCRAPING PROCESS. EACH LINE REPRESENTS AN INSTANCE FORMED BY GEOMETRIC CLASS, ADJECTIVE, NOUN, AND CONTEXTUAL MODIFIER.

Class	Adjective	Noun	Modifier	Search String
triangle	yellow	ruler	on a table	yellow+triangle+ruler+on+a+table
circle	irregular	plate	inside a container	irregular+circle+plate+inside+a+container
pentagon	rusty	road sign	side view	rusty+pentagon+road+sign+side+view

scenery, perspective, and lighting; and (iv) a noun indicating the object that represents the respective class.

Each query string was converted into a search engine-compatible format by replacing spaces with the "+" character. Furthermore, the queries were composed through controlled pseudo-random combinations, following the pattern shown in the examples in Table I; Appendix ?? details the entire process of constructing the search strings and contains the list of words used. To ensure greater diversity in the selection of parameters, mechanisms for sampling terms without replacement, random inclusion or exclusion of modifiers, and removal of duplicates were implemented.

The collection was carried out using the Chrome browser⁵, controlled by Selenium⁶ in headless mode, and the searches were performed on the DuckDuckGo search engine⁷. It was chosen for its flexibility regarding scraping, unlike others, such as Google, which block such practices. Furthermore, the chosen search engine offers impartial results, aggregates multiple sources, increasing the diversity of the collected images, and allows the use of open-license filters, which are essential for creating image datasets without bias or legal restrictions.

Each collected image was standardized to a 224x224 pixel format and converted to the RGB color space using the Pillow package⁸. This standardization was necessary to regularize the input size of the resnet50 model for feature extraction. Uniformity was ensured by an automated process, which also included the exclusion of corrupted images through exception handling. The images were organized into separate directories by geometric class (e.g., dataset/triangle/, dataset/circle/), named according to the source string and the image's index in the query (e.g., yellow_triangle_ruler_3.jpg), which facilitates their use in supervised tasks.

B. Dataset Evaluation

After the initial stage of automated image collection, the subsequent phase focuses on the quantitative assessment of the resulting dataset, as shown in Figure 2-B. The main objective of this stage is to objectively measure the quality, diversity, and consistency of the raw dataset. To this end, statistical and multivariate metrics are employed, allowing for an in-depth analysis of the intrinsic characteristics of the collected images and the initial identification of issues such as redundancies or file corruption, before proceeding to the detailed cleaning.

Before feature extraction and the application of more complex assessment metrics, the collected images went through an automatic validation pipeline. This preliminary stage aims to ensure the basic usability of the files and compliance with the defined standards. The check occurs in three main stages: (i) file integrity is verified, ensuring that each image can be correctly opened and read by image processing libraries, thus discarding truncated or corrupted files; (ii) dimensional compliance is evaluated, confirming that the dimensions of each image correspond to the pre-established 224x224 pixel standard from the collection phase; and (iii) a check is performed for the absence of visually evident corruption artifacts, such as anomalous color blocks or distortions that could compromise the analysis, also ensuring compliance with the expected RGB color space. Only the images that pass this triple validation proceed to the subsequent stages of analysis within this pipeline.

After the initial validation of the images, a deduplication process based on cryptographic hash was applied to eliminate exact redundancies in the dataset. Each validated image was converted to the RGB color space and resized to 224x224 pixels before its binary content was submitted to the SHA-256 algorithm to generate a unique identifier. The system maintains a lookup dictionary that associates each hash with its respective images, allowing for the efficient identification of duplicates through constant-time comparison $O(1)$. When images with identical hashes were detected, indicating exactly the same visual content, the pipeline preserves only the first recorded occurrence and systematically removes subsequent copies. This step was fundamental to correct cases where multiple web scraping queries retrieve the same image from different sources or alternative URLs, a common issue in web scraping practices.

Considering that the collection process was guided by specific terms for each of the nine geometric classes, the initial assessment was conducted with an intra-class focus to diagnose group-specific patterns of dispersion and redundancy. Treating each class as an independent entity at this stage is fundamental to understanding the internal cohesion and the representativeness of the images in relation to the target geometric shape. This allows for the identification of any class-specific biases or quality issues, arising from the particularities of the web scraping process for its associated terms, before performing global analyses that compare or combine the entire set of classes. This focused assessment establishes an essential baseline on the homogeneity and characteristics of each shape group.

To enable quantitative analyses, each image within a class undergoes a feature extraction process. For this purpose, the

⁵<https://www.google.com/chrome/>

⁶<https://www.selenium.dev>

⁷<https://duckduckgo.com>

⁸<https://pillow.readthedocs.io/en/stable/>

ResNet50 Convolutional Neural Network (CNN), pre-trained on the ImageNet1K V2 dataset⁹, was used, as it possesses a high capacity for generating representations rich in visual semantics [48]. Specifically, the feature vectors are typically extracted from the network’s penultimate layer, transforming each input image (pre-processed to the dimensions required by the model) into a high-dimensional numerical vector (Batch Size, 7, 7, 2048). These vectors encapsulate the relevant visual features of the image and serve as the basis for subsequent analyses of similarity, dispersion, and consistency within each class.

One of the central methods employed to assess the diversity of visual features within each class is PCA. We applied the PCA implemented in the scikit-learn package¹⁰, which employs Singular Value Decomposition (SVD) for the efficient calculation of components, to the feature vectors extracted by ResNet50, calculating the cumulative explained variance of the first 10 principal components. The choice to calculate the cumulative explained variance of the first 10 principal components is a methodological decision to establish a standardized metric, as this number proved to be empirically sufficient to capture the principal variance structure of the data and to differentiate the scenarios of interest. This analysis quantified the data dispersion in the latent space; when the variance was concentrated in the first few components (> 0.8), it indicated greater visual redundancy, whereas more uniform distributions ($[0.4 - 0.6]$) suggest greater heterogeneity and richness in the features of the images collected for that class [49]. In addition to quantifying diversity, PCA is used as a dimensionality reduction technique to facilitate the visual inspection of latent patterns.

To complement the diversity analysis, the internal cohesion of each class and the dispersion of samples in the feature space are assessed using Euclidean distances. The class centroid is calculated, which corresponds to the mean point of all feature vectors belonging to it. The Euclidean distance of each sample (image) from this centroid is then computed. The distribution of these distances, such as the mean distance to the centroid or the variance of these distances, serves as a quantitative indicator of the class’s internal cohesion: smaller values suggest a more compact clustering of samples around the mean feature, while larger values indicate greater dispersion [50]. This dispersion analysis, based on the Euclidean distances between the samples and their central point, provides a perspective complementary to that of PCA on the distribution and spread of the data in the feature space for each geometric class.

To assess structural similarity in a computationally efficient manner, SSIM is applied, whose mathematical formulation is defined in Equation 1. Unlike traditional pixel-difference-based metrics such as Mean Squared Error (MSE) and Peak Signal-to-Noise Ratio (PSNR), which only evaluate absolute errors without considering human perception, SSIM incorporates three critical components of the human visual system: luminance, contrast, and local structure [34]. This approach is

particularly relevant for sets of natural images, where subtle variations in structure can be more perceptible than intensity distortions. Due to the high volume of images per class, a random sampling approach was employed to select pairs of images for the SSIM calculation, allowing for an estimation of the class’s internal visual consistency without the need for exhaustive comparisons among all images. SSIM ranges in the interval $[-1, 1]$, with values close to 1 indicating high structural similarity [51].

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (1)$$

where:

- μ_x, μ_y are the mean pixel intensities of images x and y , respectively;
- σ_x^2, σ_y^2 represent the local variances;
- σ_{xy} is the covariance between the images;
- $c_1 = (k_1L)^2$ and $c_2 = (k_2L)^2$ are stabilization constants, where L is the dynamic range of the pixel values (e.g., 255 for 8-bit images) and $k_1 = 0.01, k_2 = 0.03$ are default values.

As demonstrated in the comparative study by [51], while PSNR shows similar values for images with different types of degradation (e.g., blurring and Gaussian noise), SSIM better discriminates these variations, aligning with subjective human assessment. For example, in the comparative study, for an image with Gaussian noise, a PSNR of 39.67 dB was associated with an SSIM of only 0.136, reflecting the high sensitivity of SSIM to structural distortions. Conversely, for compressed images, SSIM maintained high values (0.872) even with a moderate PSNR (40.39 dB), validating its ability to preserve perceived quality in non-destructive compressions.

After the individual analysis of each geometric class, the assessment pipeline consolidates the results through a statistical aggregation of the quality indicators. The intra-class metrics, including the explained variance from PCA, mean distances to the centroid, and SSIM indices, are processed to generate global descriptive statistics. The calculation of these statistics is performed using simple arithmetic means of each metric. This quantitative summary allows for verifying the overall consistency of the dataset in terms of visual diversity and structural homogeneity. It serves as an objective input for evaluating the subsequent stage of dataset outlier removal. Furthermore, the automated implementation of this process ensures the reproducibility of the assessment and provides comparable metrics for different versions of the dataset throughout the development cycle.

To facilitate the interpretation of the metrics’ results and to allow for an effective comparative analysis, the pipeline automatically generates graphical visualizations. Comparative graphics, such as boxplots, are used to illustrate the distribution of the metrics for each class, such as distances to the centroid, SSIM scores and the like, allowing the visualization of the median, the quartiles, the dispersion, and the presence of possible outliers in each group. Additionally, grouped bar charts are generated to compare mean values or other aggregated statistics of the metrics between the different classes

⁹https://www.tensorflow.org/datasets/catalog/imagenet_v2

¹⁰scikit-learn.org

side by side. These visual representations are important tools for exploratory analysis, aiding in the rapid identification of patterns, trends, and discrepancies in the quality and diversity characteristics among the evaluated geometric classes.

C. Data Cleaning

The dataset cleaning stage focuses on the identification and removal of outliers that can compromise the quality, consistency, and representativeness of the dataset for the classification task. Recognizing that different types of anomalies may be present, this phase employs a robust approach that combines three complementary outlier detection methods, each exploring a distinct perspective on the data, as shown in Figure 2-C. Techniques are used based on local density in the feature space, statistical distance measures that consider the multivariate distribution of the data, and unsupervised learning algorithms specifically designed to isolate anomalies. The union of these approaches aims to increase the reliability in identifying genuine outliers, leveraging their specificities.

The outlier detection uses as a basis the same feature vectors employed in the previous dataset assessment phase, ensuring consistency in the visual representation. As detailed in Subsection IV-B, the validated images are processed by the pre-trained ResNet50, with feature extraction from the penultimate layer before the final classification. This procedure generates for each image a 2048-dimensional vector that encapsulates its hierarchical visual features, creating a common latent space for all analyses. The reuse of these vectors, already calculated during the initial assessment stage, optimizes processing by avoiding recalculations and ensures metric coherence between the different phases of the pipeline. These normalized vectors constitute the unified input for the various outlier detection algorithms implemented in the current phase, maintaining the comparability of the results with the already established quality metrics.

Ridge regression (L2 regularization) was applied to the obtained feature vectors. This technique was used to unify the magnitude of the vectors, preventing scale differences from distorting the performance of sensitive algorithms, ensuring equitable penalization and stability in the procedure [52]. This procedure ensures that all vectors have a unified magnitude, preventing scale-sensitive algorithms from being unduly affected by the norm of the vectors. Next, the Uniform Manifold Approximation and Projection (UMAP) [53] was used, whose objective function minimizes the divergence between probability distributions in the original and reduced spaces, as described in Equation 2. This methodological choice is based on three critical aspects, detailed by [54]: (i) superiority of UMAP over PCA in a manutenção de relações topológicas complexas, particularly relevant for high-dimensional data such as the 2048-feature vectors; (ii) it is a non-linear dimensional reduction technique that preserves both the local and global structure of the data; and (iii) the ability to generate robust latent representations for subsequent neighborhood-based outlier detection.

$$\mathcal{L}_{\text{UMAP}} = \sum_{i \neq j} \left[p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) + (1 - p_{ij}) \log \left(\frac{1 - p_{ij}}{1 - q_{ij}} \right) \right] \quad (2)$$

where:

- $p_{ij} = \exp \left(-\frac{d(\mathbf{x}_i, \mathbf{x}_j) - \rho_i}{\sigma_i} \right)$ (probabilities in the original space)
- $q_{ij} = (1 + a \|\mathbf{y}_i - \mathbf{y}_j\|^{2b})^{-1}$ (probabilities in the reduced space)
- $d(\mathbf{x}_i, \mathbf{x}_j)$: distance between points in the original space
- ρ_i : minimum distance to the nearest neighbor
- σ_i : point-specific scale parameter
- a, b : curve fitting parameters (default values: $a \approx 1.93$, $b \approx 0.79$)
- $\mathbf{y}_i, \mathbf{y}_j$: coordinates in the embedding space

Among the methods used for outlier identification, HDBSCAN stands out, which is based on the density of points in the feature space and is capable of detecting clusters with varying densities. One of the main differences in relation to the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm is in the construction of a comprehensive hierarchy of clusters and the use of a stability metric to select the most relevant clusters, obviating the need to establish a global density parameter in advance. Points that do not fit into any stable cluster are considered noise and, therefore, potential outliers. A crucial aspect of the use of HDBSCAN in this study is the adaptive parameterization: the minimum cluster size hyperparameter, which affects the granularity of the detected clusters, is adjusted according to the number of samples in the class under analysis. This methodology allows the algorithm's sensitivity to adjust to the scale of each dataset, potentially increasing the precision in identifying outliers—both in classes with many samples and in those with few representatives.

Another method employed was multivariate statistical analysis based on the Mahalanobis distance, introduced in [55], for the contextual detection of outliers. This approach (Equation 3), calculates the distance of each point in relation to the center of the data distribution, taking into account the correlation between variables by means of the covariance matrix [56]. The technique was applied locally, considering the global distribution of the samples in the extracted feature space. To define the anomaly detection threshold, a density estimation of the distribution of distances was used by means of Kernel Density Estimation (KDE) [57], identifying the peak of the distribution and adding a standard deviation as the cutoff criterion [58]. Samples whose distance exceeded this threshold were classified as outliers. The advantage of this approach lies in its statistical robustness, as the Mahalanobis distance takes into account the variability and correlations among the attributes, allowing for a more precise identification of atypical points in complex multidimensional contexts.

$$D_M(\mathbf{x}_i) = \sqrt{(\mathbf{x}_i - \boldsymbol{\mu}_{\text{local}})^T \boldsymbol{\Sigma}_{\text{local}}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_{\text{local}})} \quad (3)$$

where:

- $\mathbf{x}_i$ is the point of interest

- μ_{local} is the mean of the nearest neighbors.
- Σ_{local} is the covariance matrix calculated from the nearest neighbors of $\mathbf{x}_i$.

The third method adopted for outlier removal was the Isolation Forest, first implemented in [59], responsible for isolating atypical samples by means of decision trees. Instead of calculating data density, as in neighborhood-based methods, the Isolation Forest builds multiple decision trees to partition the data recursively [60]. Samples that require fewer splits to be isolated are considered outliers, as they are more easily separable from the other samples. This method is particularly effective on high-dimensionality datasets, such as the feature vectors extracted by ResNet50 from the images, as it focuses on the efficiency of the data splitting process. To optimize the model's performance, the number of trees and the estimated proportion of outliers were dynamically adjusted, taking into consideration the quantity of images for each class. The application of Isolation Forest in the space reduced by UMAP facilitates the visualization and identification of outliers, while the choice of a dynamic threshold, based on the anomaly score, allows for a precise definition of the atypical data in an automated manner.

To combine the results of the Mahalanobis, HDBSCAN, and Isolation Forest outlier detection methods, the union of the sets of images detected by each one was considered, aiming to maximize the quantity of anomalies found. This approach ensures that the most relevant atypical samples are captured, leveraging the strengths of each approach. Furthermore, a safe removal protocol is followed, where the identified outliers are moved to an isolated directory, preserving their original meta-data for future auditing and analysis, ensuring the traceability of the process. This approach not only improves the quality of the dataset but also allows for complete and auditable documentation of the outlier detection and removal.

After the application of the outlier detection and removal methods, as detailed previously, the distribution of classes in the resulting dataset is presented in Figure 3. A considerable imbalance is observed among the classes, with some, such as "Circle" and "Ellipse," having a significantly higher number of samples compared to others, such as "Parallelogram" and "Hexagon". This imbalance is a direct consequence of the inherent difficulties in acquiring images for certain classes, reflecting the rarity or complexity of obtaining representative data in the real environment. Such disparity requires careful consideration in the choice of model evaluation metrics, aiming for a precise analysis of performance, especially for the minority classes.

D. Model Training

The training flow, described in Figure 2-D, aims to develop a classifier for the proposed case study and is structured in four main stages: data preparation, model architecture, hyperparameter optimization, and classifier training. The hyperparameter optimization and classifier training stages were carried out on a server cluster with the following available resources: two Intel Xeon Gold 6530 CPUs, totaling 24 processing cores; 64 GB of RAM; and two NVIDIA A100-SXM4-40GB GPUs,

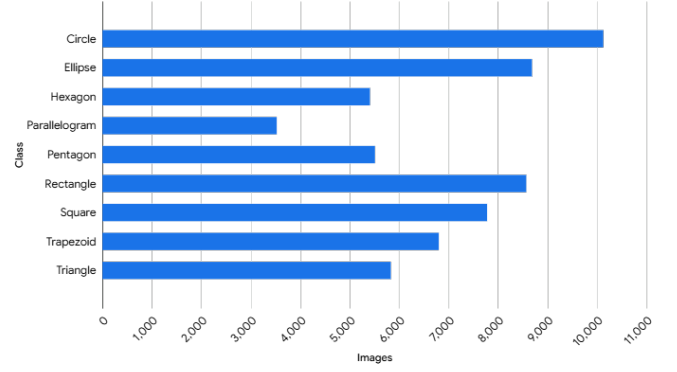


Fig. 3. Distribution of image classes after removal of outliers.

each with 40 GB of VRAM. Data preparation involved the preprocessing of images, including transformations such as resizing, normalization, and data augmentation to improve the model's robustness. In the second stage, an experimental classifier architecture is proposed, which combines the EfficientNet-B0 network, as a backbone, with an edge feature extraction module and the fusion of these features through the Efficient Channel Attention (ECA) mechanism, optimizing the network's performance ([61], [62]). The hyperparameter optimization is conducted with the use of Optuna¹¹ for the automated search of the best hyperparameters: learning rate, batch size, optimizer, scheduler, and accumulation steps. Finally, the model training and evaluation stage involved the use of gradient accumulation to handle large batches and avoids overfitting by using early stopping. During training, the model is continuously monitored, and the best model is saved based on the F1 Score evaluation. After training, the model is evaluated on the test set, and the metrics of accuracy, precision, recall, and F1 Score are calculated [63].

The generated dataset was divided in a stratified manner into train/validation and test. Next, the division of the train/validation set into training and validation parts is performed with K-Fold for cross-validation. In order to improve the model's robustness and generalization, the preprocessing of the images was performed and transformations were applied for data set augmentation. The normalization was performed based on the predefined image means and standard deviations for the EfficientNet-B0 architecture, ensuring that all images have a similar value distribution, facilitating training. Furthermore, several data augmentations are applied, including random horizontal flip, rotations of up to 10 degrees, perspective distortions, and modifications of brightness, contrast, and saturation, in addition to a Gaussian blur filter to simulate different image conditions. These transformations increased the diversity of the training images, helping the network to learn invariant features and reducing the risk of overfitting. With respect to the cross-validation strategy, a K-Fold process with $k=5$ is implemented ([64], [65]). Furthermore, the indices of the training and validation sets are stored, ensuring the reproducibility of the cross-validation process and allowing

¹¹ <https://optuna.org>(<https://optuna.org>)

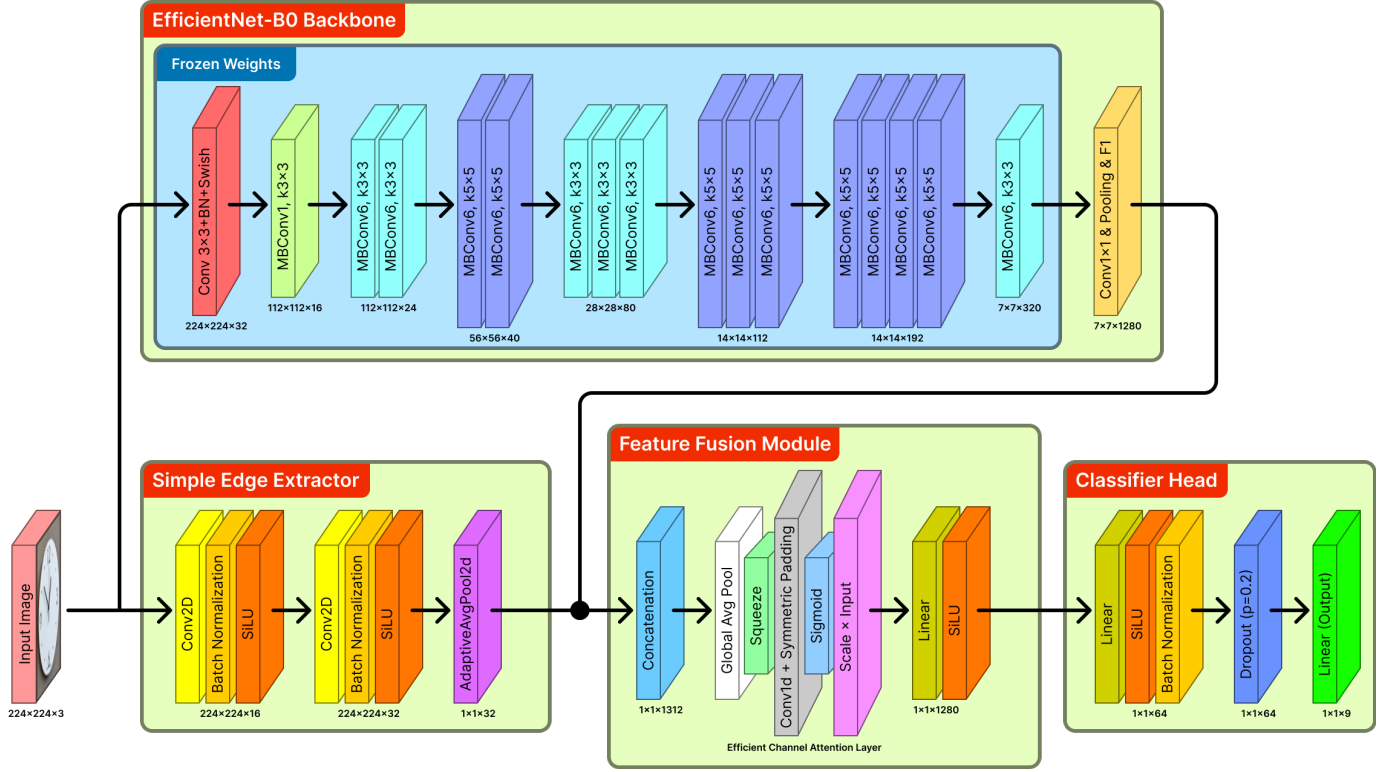


Fig. 4. Architecture of the experimental model developed for the case study. The model consists of edge extraction blocks, EfficientNet-B0 backbone, fusion of the obtained features, and a final classifier.

the results to be reproduced in future executions, guaranteeing consistency in the experiments.

The proposed model architecture, shown in Figure 4, combines the EfficientNet-B0 [66] model, a highly efficient and pre-trained convolutional neural network, with an additional Efficient Channel Attention (ECA) module, which aims to enhance the model's ability to focus on the most relevant features of the images. The EfficientNet-B0 acts as the main feature extractor, being capable of generating rich and discriminative representations of the input images. To complement, an edge feature extractor is integrated, which concentrates on capturing crucial details of the images' geometric structure, using lightweight convolutional layers to preserve computational efficiency. The fusion of these extracted features is performed by means of a feature fusion mechanism via weighted concatenation, in which the representations from EfficientNet-B0 and the edge extractor are combined in a way to maximize the relevant information, while the ECA module dynamically adjusts the contribution of each set of features. The use of ECA is particularly effective, as it applies channel-wise attention, highlighting the most relevant features in an efficient manner, while the weighted concatenation allows the model to learn the best combined representations, improving its generalization and accuracy capability.

The search for the best hyperparameters is conducted with the use of Optuna. For each fold, the model is trained using the training set and evaluated on the validation set. The performance metrics, such as F1 Score, are calculated for each fold, and the fold with the best performance is selected as

the best fold. This cross-validation approach ensures that the model is robustly evaluated on different subsets of the data, avoiding overfitting and ensuring better generalization of the model. The test set is kept separate and is not used in the cross-validation process, ensuring that the final evaluation of the model is done in an unbiased manner. After finding the best fold within the train/validation set, the model trained on that fold is used for final hyperparameter optimization and evaluation on the test data.

The learning rate is optimized in a continuous space, varying from 1×10^{-6} to 1×10^{-2} , with a logarithmic distribution to explore a wide range of values, from low to higher rates, adjusting to the model's need during training. For categorical parameters, the batch size is optimized in a set of discrete values, including 64, 128, 256, 384, 512, allowing adjustment of the processing capacity and training efficiency. Furthermore, two types of optimizers are considered, Adam and AdamW, which are chosen depending on the problem's characteristics and the model's behavior, directly impacting the weight updates. Other optimized hyperparameters include the number of gradient accumulation steps, varying from 1 to 8, and the weight decay, adjusted between 1×10^{-6} and 1×10^{-2} to control overfitting. These options were integradas ao processo de validação cruzada and, at each iteration, the model was trained and evaluated with the chosen hyperparameters, seeking to maximize the F1 Score metric.

The model training was conducted in a structured manner, with a focus on the efficiency and robustness of the learning. During training, the model was fed with the training data

by means of optimized DataLoaders, using the best set of hyperparameters found in the optimization stage. At each epoch, the model underwent a gradual update of its weights, adjusting based on the cross entropy loss function, while the selected optimizer performed the updates on the parameters. Gradient accumulation was applied to handle large batches without overloading the memory, allowing the gradients to be accumulated and applied at defined intervals [67]. Furthermore, the training is complemented with a learning rate scheduler, dynamically adjusting the learning rate throughout the epochs to optimize the convergence process [68]. To prevent overfitting, techniques such as early stopping and checkpointing were used, interrupting the training if the model does not improve after a defined number of epochs and saving the best model based on the F1 Score metric. During the process, the performance metrics, such as F1 Score, accuracy, and precision, are monitored, and the model is validated after each epoch to ensure there is no performance degradation, maximizing the model's generalization to unseen data.

The model performance evaluation was carried out with an emphasis on the weighted F1-Score, which was adopted as the main evaluation metric due to its ability to balance precision and recall, especially in scenarios with imbalanced classes. The F1-Score provides a robust measure of performance, as it takes into account both false positives and false negatives, being particularly useful and judicious in classification problems where the minimization of errors in both classes is crucial [69]. Furthermore, to provide a more comprehensive view of the model's efficiency, other complementary metrics were calculated, such as Giga Floating Point Operations Per Second (GFLOPs), which estimates the computational cost of the model in terms of floating-point operations per second, offering a measure of its computational efficiency.

The inference time was also monitored, allowing an analysis of the model's speed in making predictions, which is essential in production environments or real-time applications. The confusion matrix was analyzed for each class, allowing a detailed understanding of the model's assertiveness in correctly identifying the different categories. This analysis was complemented by the generation of a classification report that details the precision, recall, and F1-Score per class, allowing the identification of areas where the model can be improved. These metrics provide a complete evaluation of the model, considering not only accuracy but also its efficiency and behavior in each specific class, ensuring a balanced view between performance and computational cost.

V. RESULTS AND DISCUSSION

The data collection was carried out based on the structured search data. 6000 search strings were generated, from which it was possible to capture 106,048 images in total across the 9 classes established for the case study. Each geometric class was associated with a list of everyday objects that typically manifest that shape, totaling 191 objects distributed among the 9 distinct geometric classes. For each query performed, the first 50 images generated on the query page were selected, although several queries provided a lower number of images.

The generated dataset and its features are available on Hugging Face¹².

The outlier removal was performed using three distinct detection techniques: Mahalanobis, HDBSCAN, and Isolation Forest. Table II presents the quantity of outliers detected for each class, as well as the total of outliers identified by each technique. On average, the Mahalanobis method identified the highest number of outliers, with 4528.14 outliers per class, followed by HDBSCAN, which detected an average of 2030.26 outliers, and Isolation Forest, with 1444.82 outliers detected per class. The total of outliers per class was calculated from the union of the results of the three methods, and the general average of outliers for all classes was 5883.85, with a standard deviation of 1444.82, indicating a considerable dispersion in the data. This multi-method approach ensured a more robust identification of atypical samples, with the objective of improving the dataset's quality, maintaining its representativeness and consistency for the subsequent phases of the model. The plotting of the three outlier detection techniques provided a clear and intuitive visualization of the atypical samples identified by each method. Figure 5 presents an example of the dispersion graphs for the parallelogram class, which illustrate the distribution of the samples in the reduced space.

The removal of outliers demonstrated a quantifiable impact on the quality of the dataset for the proposed case study, as evidenced by the comparative metrics displayed in Figure 6. In the initial state, the dataset presented a mean PCA variance of 0.2586, a mean distance to the centroid of 11.8642 (± 1.9691), and a mean SSIM of 0.2971 (± 0.1736), indicating significant dispersion and low intra-class structural consistency. After the filtering process, an increase of 25.6% in the mean PCA variance (0.3247) was observed, suggesting greater diversity of valid features. The 5.6% reduction in the mean distance to the centroid to 11.1957 (± 1.7291) reflects a more cohesive distribution of the samples in the latent space, while the 21.6% increase in the mean SSIM (0.3613 ± 0.1891) attests to the greater visual homogeneity of the classes. These results align with previous studies [51], in which SSIM values much lower than 0.4 were associated with perceptual inconsistencies. The concomitant decrease in the standard deviation of the distance to the centroid (-12.2%) corroborates the effectiveness of the method in eliminating deviant samples without compromising legitimate variability.

The hyperparameter optimization process was conducted over 100 automatic searches using cross-validation with 5 folds. The objective function was defined to maximize the F1-Score on the validation set, ensuring a balance between precision and recall in scenarios with imbalanced classes. At the end of the process, the best hyperparameters found were: batch size of 256, learning rate of approximately 0.0032, AdamW optimizer, StepLR scheduler, accumulation steps equal to 2, and weight decay of approximately $6,21 \times 10^{-4}$. These values were used in the final model training stage and served as the basis for selecting the best fold, subsequently evaluated on the test set. The choice of AdamW and the

¹²(<https://huggingface.co/datasets/Horusprg/WildShapes>)

TABLE II

QUANTITY OF OUTLIERS IDENTIFIED PER CLASS WITH THE THREE EMPLOYED METHODS: MAHALANOBIS, HDBSCAN, AND ISOLATION FOREST. THE LAST COLUMN PRESENTS THE TOTAL OF UNIQUE OUTLIERS IDENTIFIED PER CLASS AFTER THE UNION STRATEGY. THE MEAN VALUES AND STANDARD DEVIATIONS PER METHOD ARE HIGHLIGHTED IN THE LAST ROW.

Classes	Mahalanobis	HDBSCAN	Isolation Forest	Total Outliers
circle	4583	1186	1623	6092
ellipse	3888	22	1354	4846
hexagon	2503	611	880	3379
parallelogram	3037	4033	944	5905
pentagon	2982	278	927	3750
rectangle	5006	110	1462	6041
square	4437	95	1304	5256
trapezoid	3280	323	1108	4275
triangle	3447	0	1007	4227
mean ($\pm$ std)	4528.14 ($\pm$ 843.36)	2030.26 ($\pm$ 1290.48)	1444.82 ($\pm$ 266.04)	5883.85 ($\pm$ 1444.82)

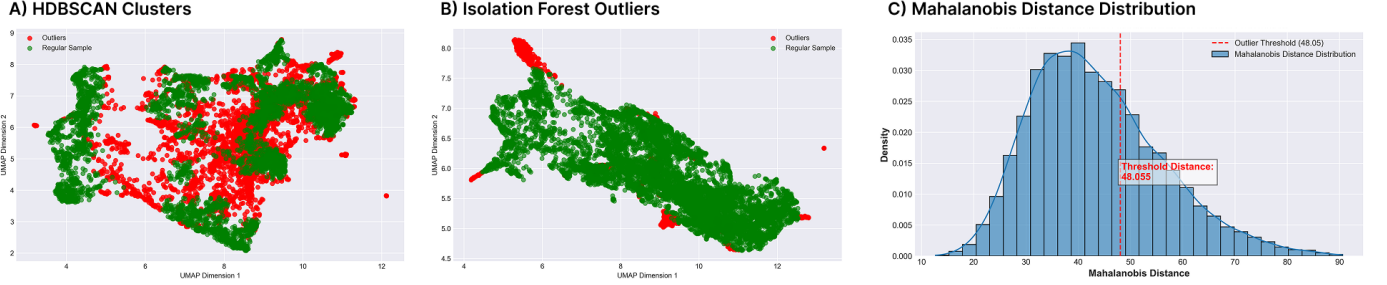


Fig. 5. Example of graphs generated for each outlier removal method for the parallelogram class. (A) Analysis by HDBSCAN, identifying 4,033 outliers. (B) Analysis by Isolation Forest, detecting 944 outliers. (C) Analysis by Mahalanobis Distance, with the threshold set at 48.055, identifying 3,037 outliers. Graphs (A) and (B) use the UMAP 1 (x-axis) and UMAP 2 (y-axis) dimensions, while graph (C) represents the Mahalanobis Distance (x-axis) versus Density (y-axis).

StepLR scheduler indicates that the model benefited from a more stable regularization and a decreasing learning rate in discrete steps, adapting well to the proposed scenario.

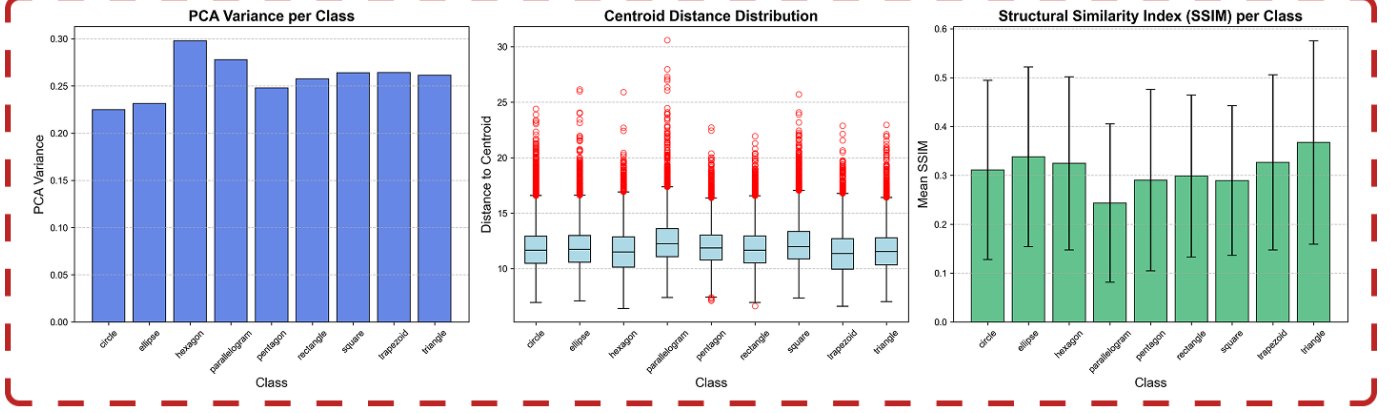
Figure 7 presents a visual analysis of the mean performance obtained for each value range of the hyperparameters explored during optimization. It is observed that the best F1-Score results occurred mostly when the learning rate was in the range of 10^{-3} , and the weight decay concentrated the best performances between the ranges of 10^{-4} and 10^{-3} , balancing well the penalty for complexity without affecting convergence. The optimal batch size was 256, providing stability without compromising training speed. Furthermore, the folds that most frequently resulted in good performances were the intermediate ones (between folds 1 and 4), and the use of 4 accumulation steps also stood out in the combinations with the highest mean F1-Score. This analysis highlights the importance of a well-distributed search in the hyperparameter space and reinforces the impact of the careful choice of these values on the final performance of the model.

During the training process of the developed model, a

consistent evolution of performance was observed, as illustrated in Figure 8. The F1-Score metric was used to track the performance on both the training and validation sets, revealing a stable trend of learning and generalization. The difference between the training and validation curves remained controlled throughout the epochs, which indicates that the model did not suffer from significant overfitting. This behavior suggests that the adopted strategies, such as the use of regularization via weight decay, dropout, and gradient accumulation, were effective in ensuring the robustness of the learning process, even with the focus on computational efficiency ([70], [71]).

Figure 9 presents the variation of the learning rate during the training epochs, conducted with the StepLR scheduling policy. This policy promotes periodic reductions of the learning rate, allowing the model to make fine adjustments in the last stages of training. Initially, the higher learning rate facilitates rapid convergence, while the subsequent reductions favor stability in the approximation of local minima. The controlled variation of the learning rate was essential to achieve a balance between convergence speed and final model quality [72], contributing

Initial Dataset



Cleaned Dataset

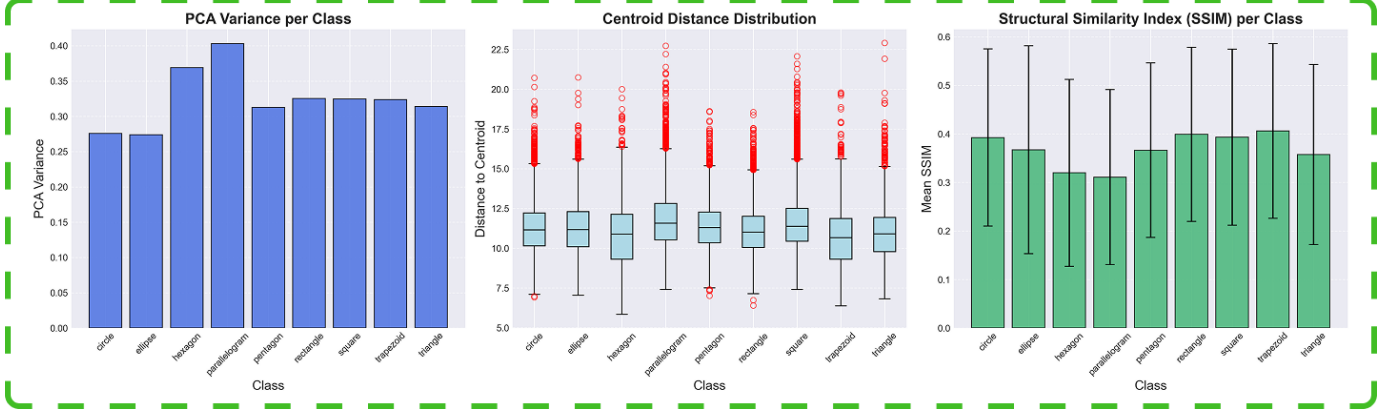


Fig. 6. Comparison between the initial generated dataset and after the removal of outliers, considering the PCA variance per class, the distribution of the centroids found per class, and the SSIM per class.

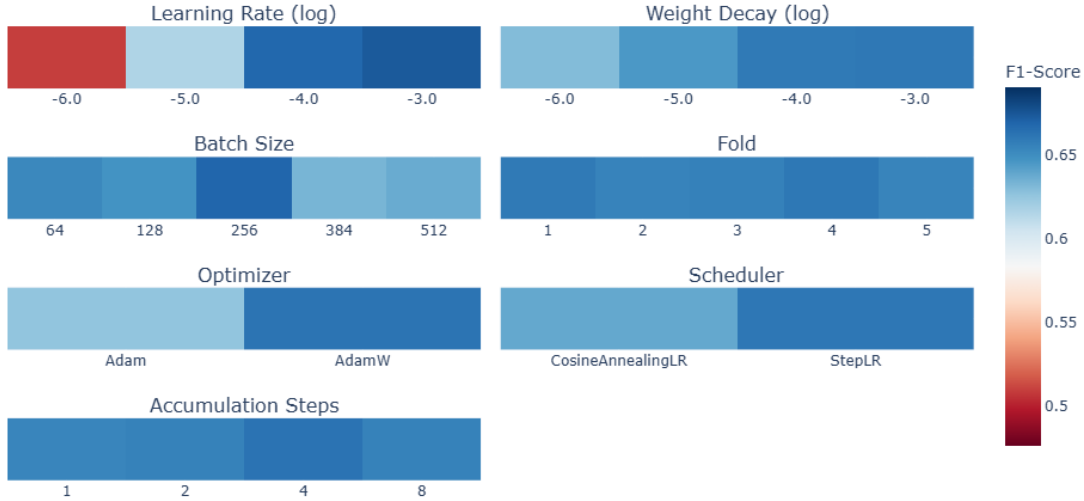


Fig. 7. Mean F1-Score obtained for each value range of the selected hyperparameters for the classifier optimization.

to the good results observed in the evaluation stage.

During the final evaluation stage, the classification model achieved an expressive performance on the test set, with an F1-Score of 92.5%, an accuracy of 92.5%, and a mean inference time of approximately 0.49 seconds per image, reflecting a rate of 2.02 inferences per second. The computational cost mea-

sured in GFLOPs was approximately 0.64, which indicates a processing-efficient model, suitable for applications on devices with limited computational resources. Figure 11 presents the confusion matrix with the correct predictions and errors per class, evidencing a balanced accuracy rate among the different geometric shapes. Figure 10, in turn, illustrates the complete

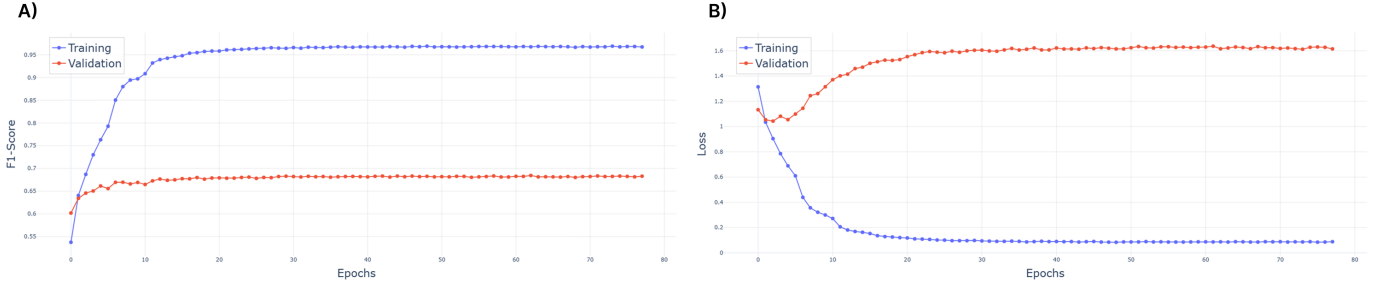


Fig. 8. F1-Score obtained on the training and validation sets during the training of the developed experimental model.

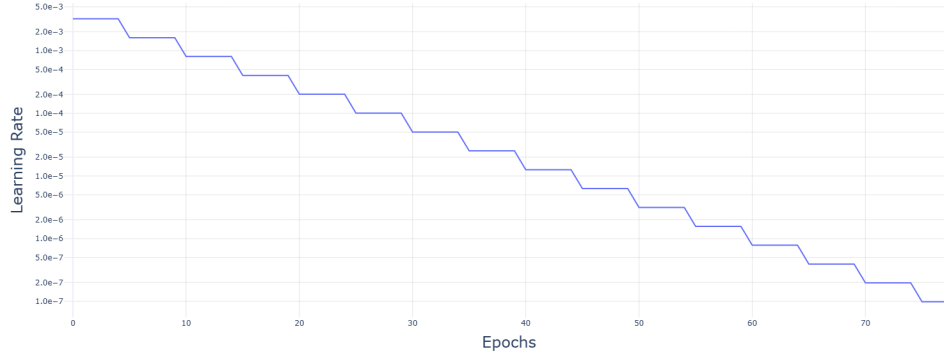


Fig. 9. Variation of the learning rate throughout the training epochs of the classifier.

report of the performance metrics per class, with precision, recall, and F1-score values varying between 90% and 95%, which reinforces the model's robustness even in multi-category scenarios.

Despite the positive results, a significant discrepancy was observed between the performance obtained during validation (mean F1-Score of 68%, on the red curve of Figure 8-A) and the final results on the test set. A possible explanation for this difference lies in the absence of stratification during the data separation process for validation. Without stratification, the distribution of classes among the folds may have been imbalanced, leading the model to be evaluated in scenarios non-representative of the dataset's real distribution. This may have caused an underestimation of the performance during the validation phase, masking the model's real capacity to generalize. The use of stratified validation in future works concerning the pipeline will be implemented, in order to ensure a more faithful assessment of the original class distribution.

Considering the results obtained, the proposed method presented a consistent and effective performance in the context of the case study, aimed at supporting the GeoMeta educational application. The automated dataset curation stage, combined with the removal of outliers by consensus among three distinct techniques, resulted in a cleaner and more representative dataset, which was directly reflected in the quality of the trained model. The final precision of 92.5% on the test set, accompanied by an F1-score with a similar value, evidenced the assertiveness of the model built based on the refined dataset. Furthermore, the significant reduction in variance between classes and the control over noisy samples contributed

to a more robust generalization. Such results reinforce the applicability of the method in real scenarios of low connectivity and infrastructure, especially due to the low GFLOP cost, demonstrating its capacity to produce efficient models in the Amazonian context from automated processes of visual data collection and treatment.

VI. CONCLUSION

This work aimed to develop an automated and structured method for the construction, curation, and evaluation of visual datasets aimed at the classification of two-dimensional geometric shapes, with a focus on low-computational-cost educational applications. Based on the proposed methodology, the following stages were conducted: (i) automated extraction of visual features from images by means of a lightweight and efficient architecture; (ii) application of an outlier detection and removal protocol using multiple unsupervised approaches (Mahalanobis, HDBSCAN, and Isolation Forest) with a consensus strategy to maximize the identification of anomalies; (iii) organization and structuring of a specific dataset for the task of classifying two-dimensional geometric shapes, targeting educational scenarios with low connectivity; and (iv) implementation of a complete flow of hyperparameter optimization with stratified cross-validation, training, and evaluation of models with a focus on computational cost.

Regarding the results obtained with the produced model, the performance achieved on the test set stands out, with 92.5% accuracy and F1-score, in addition to a reduced computational cost, measured by means of only 0.63 GFLOPs and a mean rate of 2 inferences per second. These results demonstrate that

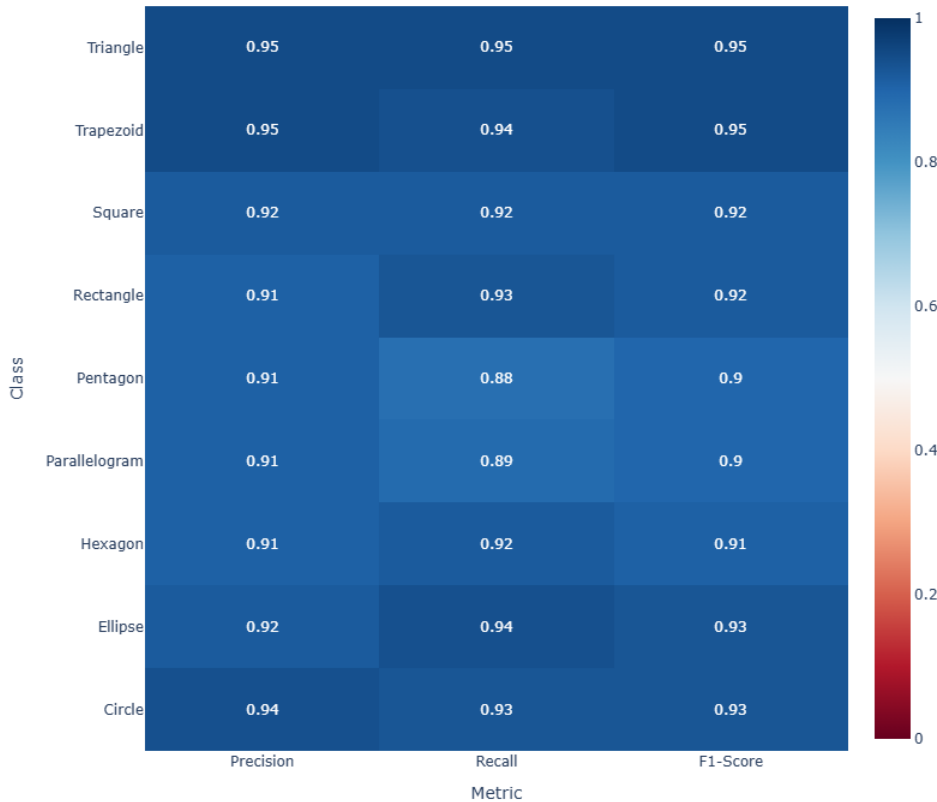


Fig. 10. Results obtained by the trained model on each class in the test set for the metrics precision, recall, and F1-Score.

the adopted methodological flow allowed for the construction of a robust and efficient model, even in the face of the hardware and connectivity restrictions foreseen in the case study. The analysis of the confusion matrix and the per-class report evidenced a good discriminative capacity among the nine geometric categories, confirming the feasibility of the application in real-use scenarios.

Based on these results, the method demonstrated its capability to produce, in an efficient and automated manner, a clean and representative dataset, in addition to allowing the training of models compatible with offline execution on mobile devices. The case study focused on the GeoMeta application reinforced the practical applicability of the proposal, evidencing its usefulness in basic education contexts in the Amazon, where there are technical and structural restrictions for the use of conventional AI solutions.

Among aspects to be improved in this work, the absence of specific public benchmarks for the task of classifying two-dimensional geometric shapes in everyday objects stands out, which makes direct comparisons with other approaches difficult. Furthermore, although a cross-validation protocol was adopted, the performance variation between the folds suggests that the class distribution in the train/validation set can still be refined. Finally, the model, although lightweight,

still requires a device with minimum graphics processing capability, which may limit its use on extremely basic or old devices. Another limitation of this work lies in the removal of duplicate images in the evaluation stage of the generated dataset, as cryptographic hashes are sensitive to minimal changes; therefore, to identify approximate variants or visually very similar images, which are not detected by this method, complementary techniques would be necessary, such as those based on perceptual hashes or on the similarity analysis of the extracted feature vectors.

As future work, it is proposed to expand the presented method beyond image classification, exploring its application to different types of data, such as texts and tabular bases, which can significantly broaden its impact in interdisciplinary scenarios. Another promising avenue is the adaptation of the method for more complex tasks, such as object detection and semantic segmentation, which will require adjustments in feature extraction and in automated visual validation. Furthermore, it is intended to investigate the integration of new outlier removal and active learning algorithms to prioritize ambiguous or representative samples, further optimizing the automatic curation process. Finally, the creation of an interactive interface for continuous monitoring and refinement of the curation—by specialists or end-users can increase the

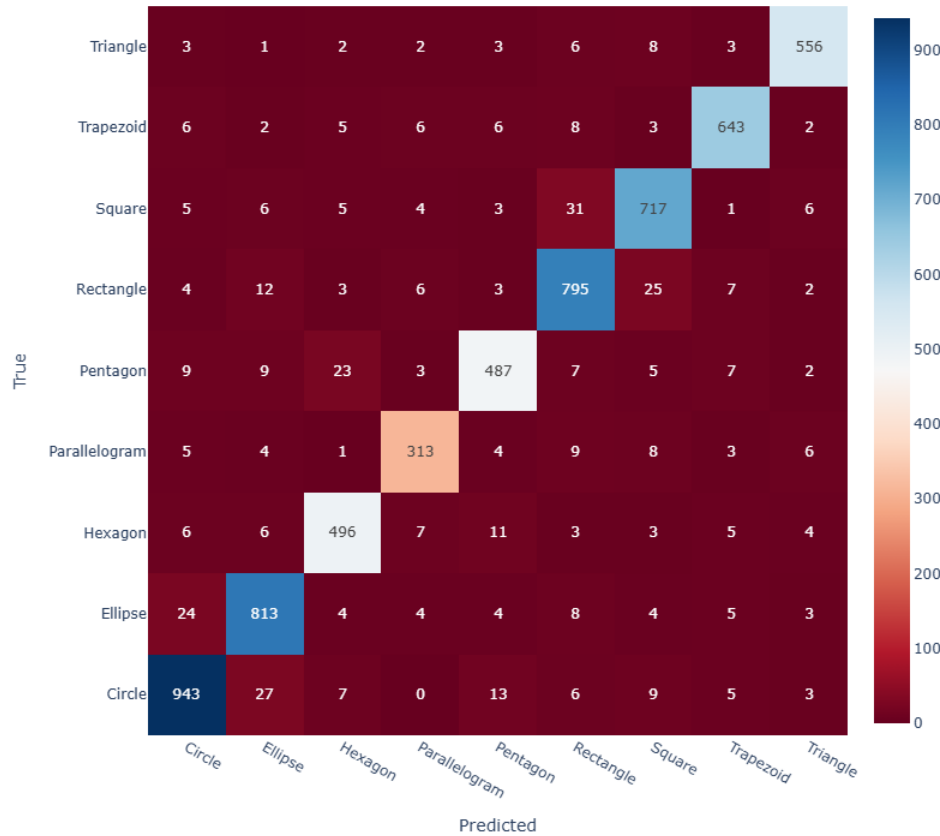


Fig. 11. Confusion matrix comparing the true class with the one predicted by the classifier on the test set.

reliability of the generated dataset in critical domains.

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